Chap. 1. Signals and Systems

§1.3 Exponential and Sinusoidal Signals

§1.3.1 Continuous-time Complex Exponential & Sinusoidal Signals

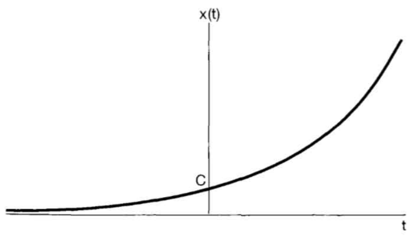
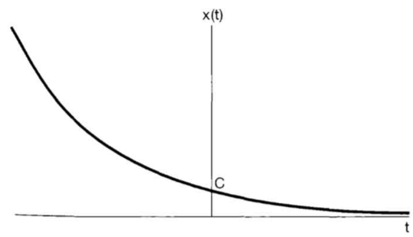
- Continuous-time complex exponential signal

$$x(t) = Ce^{at} \qquad (1.20)$$

C and a are, in general, complex numbers.

(1) Real Exponential Signals

If C and a are real, $x(t)$ is called a real exponential.

$a > 0$ (a growing exponential)	$a < 0$ (a decaying exponential)
	
- chain reactions in atomic explosions - chemical reactions	- the process of radioactive decay - the responses of RC circuits - damped mechanical systems

If $a = 0$, then $x(t) =$ constant.

(2) Periodic Complex Exponential and Sinusoidal Signals

If C is real and a is purely imaginary, $x(t)$ is sinusoidal.

Consider a periodic complex exponential signal

$$x(t) = e^{j\omega_0 t}$$

This signal will be periodic with period T, if

$$e^{j\omega_0 t} = e^{j\omega_0 (t+T)} = e^{j\omega_0 t} e^{j\omega_0 T}$$

(3) General Complex Exponential Signals

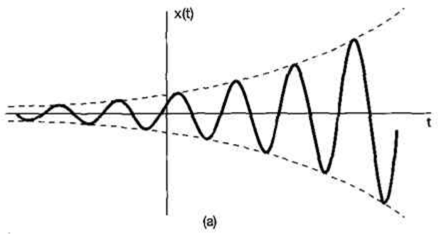
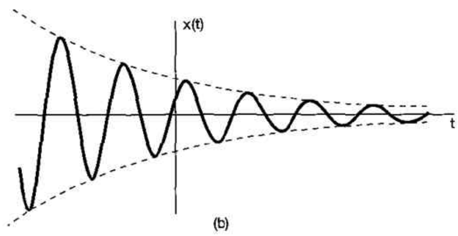
- Continuous-time complex exponential signal

$$x(t) = C e^{at} \qquad (1.20)$$

C and a are, in general, complex numbers.

The complex numbers C and a are represented as following:

$C = |C| e^{j\theta}$ and $a = r + j\omega_0$

growing sinusoidal signal	decaying sinusoidal signal
$x(t) = C e^{rt} \cos(\omega_0 t + \theta),\ r > 0$	$x(t) = C e^{rt} \cos(\omega_0 t + \theta),\ r < 0$
 (a)	 (b)
※ unstable systems (The poles of a system are on the right half s-plane.)	the response of RLC circuits the automotive suspension systems

※ <u>Summary</u>

$$x(t) = C e^{at}$$

C	a	example	real part
real	real	$5e^{-2t}$	$5e^{-2t}$
real	complex	$5e^{(-2+j3)t}$	$5e^{-2t}\cos(3t)$
complex	real	$(5+j5)e^{-2t}$	$5e^{-2t}$
complex	complex	$(5+j5)e^{(-2+j3)t}$	$5\sqrt{2}e^{-2t}\cos(3t+45^\circ)$
(*) real	pure imaginary	$5e^{j3t}$	$5\cos(3t)$
(*) real	0	$5e^{0t}$	5

§1.3.2 Discrete-time Complex Exponential and Sinusoidal Signals

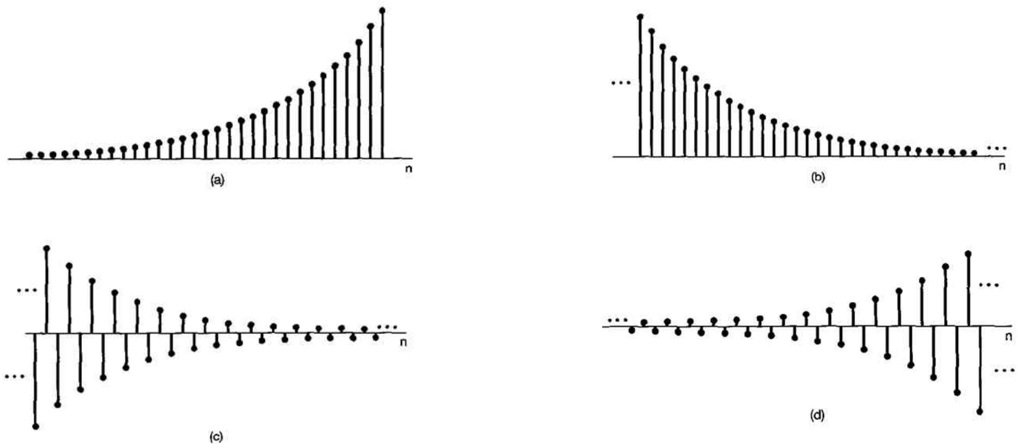
- Discrete-time complex exponential signal (or sequence)

$$x[n] = C\alpha^{n} \quad \text{or} \quad x[n] = Ce^{\beta n} \qquad (1.44)$$

C and α are, in general, complex numbers.

(1) Real Exponential Signals

$x[n] = C\alpha^{n}$, $(a)\ \alpha > 1$; $(b)\ 0 < \alpha < 1$; $(c)\ -1 < \alpha < 0$; $(d)\ \alpha < -1$



(2) Periodic Complex Exponential and Sinusoidal Signals

β is purely imaginary

$$x[n] = e^{j\omega_0 n}$$

Periodic Complex Exponential and Sinusoidal Signals

	periodic complex exponential signal	sinusoidal signal
discrete-time	$x[n] = e^{j\omega_0 n}$	$x[n] = A\cos(\omega_0 n + \phi)$
continuous-time	$x(t) = e^{j\omega_0 t}$	$x(t) = A\cos(\omega_0 t + \phi)$

(3) General Complex Exponential Signals

- Discrete-time complex exponential signal (or sequence)

$$x[n] = C\alpha^n \quad \text{or} \quad x[n] = Ce^{\beta n} \tag{1.44}$$

C and α are, in general, complex numbers.

The complex numbers C and a are represented as following:

$$C = |C|\, e^{j\theta} \quad \text{and} \quad \alpha = |\alpha|\, e^{j\omega_0}$$

Then,

$$x[n] = C\alpha^n = |C|\,|\alpha|^n \cos(\omega_0 n + \theta) + j\,|C|\,|\alpha|^n \sin(\omega_0 n + \theta)$$

$|\alpha| = 1$: sinusoidal

$|\alpha| < 1$: sinusoidal sequences multiplied by a decaying exponential

$|\alpha| > 1$: sinusoidal sequences multiplied by a growing exponential

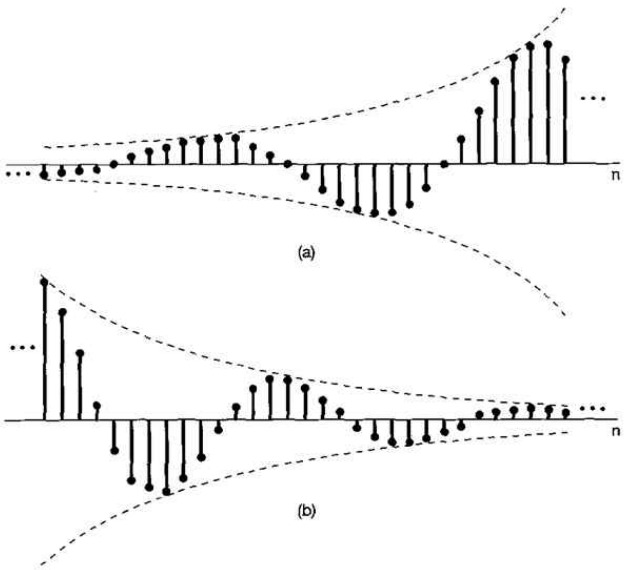


Figure 1.26 (a) Growing discrete-time sinusoidal signals; (b) decaying discrete-time sinusoid.

§1.3.3 Periodicity Properties of Discrete-time Complex Exponentials

While there are many similarities between continuous-time and discrete-time signals, there are also many important differences.

● One of these concerns the discrete-time exponential signal $e^{j\omega_0 n}$.

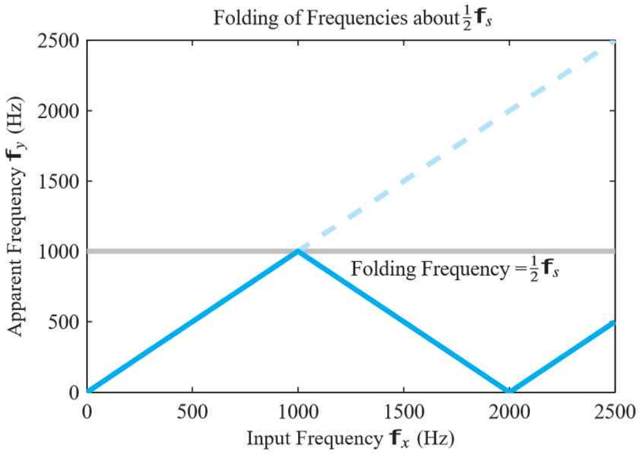
continuous-time signal $e^{j\omega_0 t}$	discrete-time signal $e^{j\omega_0 n}$
$\omega_0 \uparrow \Rightarrow$ the rate of oscillation $\uparrow$	?
periodic for any value of ω_0	?

Right or Wrong?

discrete	$e^{j(\omega_0+2\pi)n} = e^{j2\pi n} e^{j\omega_0 n} = 1\ e^{j\omega_0 n} = e^{j\omega_0 n}$	$e^{j2\pi n} = 1$?
continuous	$e^{j(\omega_0+2\pi)t} = e^{j2\pi t} e^{j\omega_0 t} = 1\ e^{j\omega_0 t} = e^{j\omega_0 t}$	$e^{j2\pi t} = 1$?

※ $\omega_0 \uparrow \Rightarrow$ the rate of oscillation $\uparrow$ in the continuous-time domain?

※ $\omega_0 \uparrow \Rightarrow$ the rate of oscillation $\uparrow$ in the discrete-time domain?



sampling frequency = 2000 Hz

※ Physical meaning of ω_0 [rad/sec] in the continuous-time domain?

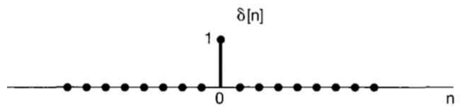
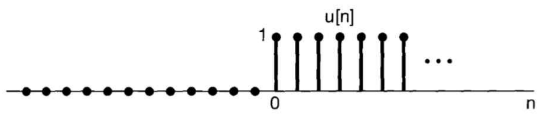
※ Physical meaning of ω_0 [rad] in the discrete-time domain?

continuous-time signal $e^{j\omega_0 t}$	ω_o [rad/sec] **is the real frequency.**
discrete-time signal $e^{j\omega_0 n}$	ω_o [rad] **is not the real frequency, but the sampling interval in radian.**

1.4 The Unit Impulse and Unit Step Functions

1.4.1 The Discrete-time Unit Impulse and Unit Step Sequences

- Basic discrete-time signals

discrete-time unit impulse (or unit sample)	discrete-time unit step
$\delta[n] = \begin{cases} 0, & n \neq 0 \\ 1, & n = 0 \end{cases}$	$u[n] = \begin{cases} 0, & n < 0 \\ 1, & n \geqq 0 \end{cases}$
 Fig. 1.28	 Fig. 1.29

1.6 Basic System Properties

1.6.1 Systems with and without Memory

1.6.2 Invertibility and Inverse Systems

1.6.3 Causality

1.6.4 Stability

1.6.5 Time Invariance

1.6.6 Linearity

1. Homogeneity
2. Additivity

(homogeneity + additivity)

Continuous time : $a x_1(t) + b x_2(t) \rightarrow a y_1(t) + b y_2(t)$

Discrete time : $a x_1[n] + b x_2[n] \rightarrow a y_1[n] + b y_2[n]$

Chap. 2. Linear Time Invariant System

● Linear Time-Invariant Systems

- Superposition property (ex: circuit theory)

a signal $\Rightarrow$ LTI system $\Rightarrow$ the output

$\Downarrow$ $\Uparrow$

a linear combination of a set of basic signals $\Rightarrow$ LTI system $\Rightarrow$ the linear combination of its responses to basic signals

2.1 Discrete-time LTI systems

2.1.1 Discrete-time signals in terms of impulses

- Weighted time-shift impulses

$$x[-1]\delta[n+1]=\begin{cases} x[-1], & n=-1 \\ 0, & n\neq -1 \end{cases}$$

$$x[0]\delta[n]=\begin{cases} x[0], & n=0 \\ 0, & n\neq 0 \end{cases}$$

$$x[1]\delta[n-1]=\begin{cases} x[1], & n=1 \\ 0, & n\neq 1 \end{cases}$$

- More generally, $x[n]$ can be written:

$$\begin{aligned} x[n] &= \cdots + x[-2]\delta[n+2]+x[-1]\delta[n+1] \\ &\quad + x[0]\delta[n]+x[1]\delta[n-1]+x[2]\delta[n-2]+\cdots \\ &= \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \qquad (2.2) \end{aligned}$$

· Decomposition of a discrete-time signal into a weighted sum of shifted impulses

- Eq(2.2): SIFTING property (NOT shifting property)
  An arbitrary signal can be represented as a linear combination of shifted unit impulses $\delta[n-k]$, where the weights are $x[k]$.

※ [Importance of Eq.(2.2) $x[n]=\sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$]

- Any discrete-time signal can be represented by using Eq. (2.2).
  [i.e,, the weighted sum of time-shifted impulses]
- The impulse is the basic element of signals.
- A signal can be represented
  a summation [※ linear (additivity)]
  of the weighted [※ linear (homogeneity)]
  time-shifted impulses. [※ time-invariant]
- linear combination (i.e., a kind of series)

5/2

2.1.2 The convolution sum representation of LTI systems

- **IMPULSE RESPONSE**

  the output of a system $h[n]$, when the impulse $\delta[n]$ is applied to the system as the input

impulse $\delta[n]$ → system H → $y[n]=h[n]$

- the **importance** of the impulse response

  If we know the impulse response of an LTI system, we can get the response (i.e., output) of the system for any input by using Eq. (2.2) and Eq. (2.6).

- (Given) the impulse response of an LTI system : $h[n]$

$$\delta[n] \quad \Rightarrow \quad h[n] \qquad \text{(known)}$$

$$\begin{array}{rcll} \cdots & \Rightarrow & \cdots & \\ \delta[n] & \Rightarrow & h[n] & \text{(known)} \\ \delta[n-1] & \Rightarrow & h[n-1] & \text{(time invariant)} \\ \delta[n-2] & \Rightarrow & h[n-2] & \text{(time invariant)} \\ \cdots & \Rightarrow & \cdots & \end{array}$$

$$\begin{array}{rcll} \cdots & \Rightarrow & \cdots & \\ a_0\delta[n] & \Rightarrow & a_0 h[n] & \text{(linear)} \\ a_1\delta[n-1] & \Rightarrow & a_1 h[n-1] & \text{(linear)} \\ a_2\delta[n-2] & \Rightarrow & a_2 h[n-2] & \text{(linear)} \\ \cdots & \Rightarrow & \cdots & \end{array}$$

$$\begin{aligned} &\cdots+\delta[n]+\delta[n-1]+\delta[n-2]+\cdots \\ &\qquad\qquad \Rightarrow \quad \cdots+h[n]+h[n-1]+h[n-2]+\cdots \qquad \text{(linear)} \end{aligned}$$

$$\begin{aligned} &\cdots+a_0\delta[n]+a_1\delta[n-1]+a_2\delta[n-2]+\cdots \\ &\qquad\qquad \Rightarrow \quad \cdots+a_0 h[n]+a_1 h[n-1]+a_2 h[n-2]+\cdots \qquad \text{(linear)} \end{aligned}$$

$$\begin{aligned} x[n] &= \cdots+a_0\delta[n]+a_1\delta[n-1]+a_2\delta[n-2]+\cdots \\ &= \cdots+x[0]\delta[n]+x[1]\delta[n-1]+x[2]\delta[n-2]+\cdots \\ &= \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \qquad \text{(linear combination of shifted impulses)} \end{aligned}$$

$$\begin{aligned} y[n] &= \cdots+a_0 h[n]+a_1 h[n-1]+a_2 h[n-2]+\cdots \\ &= \cdots+x[0]h[n]+x[1]h[n-1]+x[2]h[n-2]+\cdots \\ &= \sum_{k=-\infty}^{\infty} x[k]h[n-k] \qquad \text{(convolution sum)} \end{aligned}$$

$$\boxed{\begin{aligned} x[n] &= \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \qquad \text{(sifting property)} \qquad (2.2) \\ y[n] &= \sum_{k=-\infty}^{\infty} x[k]h[n-k] \qquad \text{(convolution sum)} \qquad (2.6) \end{aligned}}$$

- the symbol of convolution

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] \equiv x[n] * h[n] \qquad (2.7)1$$

	discrete-time	continuous-time
sifting property	$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$	$x(t) = \int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau)\,d\tau$
convolution sum/integral	$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$	$y(t) = \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau$
	Eq.(2.2) / Eq.(2.6)	Eq.(2.27) / Eq.(2.33)

§2.6 Summary

- system representation methods in the time domain
  1. impulse response
  2. linear constant-coefficient differential equation
  3. block diagram
  4. state-variable description
- sifting property
  weighted sums [or integrals] of shifted unit impulses
- convolution sum [or integral]
- The sifting and convolution properties are extremely important, as they allow us to compute the response of an LTI system to an arbitrary input in terms of the system's response to a unit impulse.

Chap 3. Fourier Representations for Periodic Signals

- Continuous-Time Periodic Signals
- Continuous-Time Fourier Series (CTFS)
- Discrete-Time Periodic Signals
- Discrete-Time Fourier Series (DTFS)
- Gibbs Phenomenon
- Properties of Fourier Representation

- Discrete-Time Nonperiodic Signals
- Discrete-Time Fourier Transform (DTFT)
- Continuous-Time Nonperiodic Signals
- Continuous-Time Fourier Transform (CTFT)

※ Can we make a square wave by adding sinusoidal waves? [YES !!]

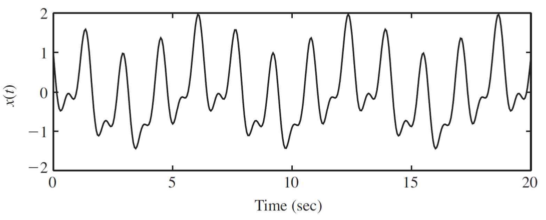
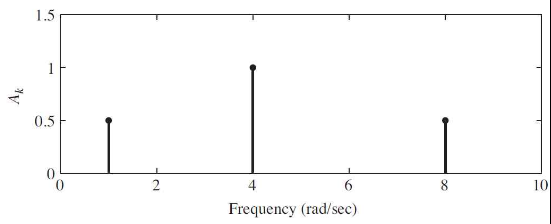
§3.2 The response of LTI systems to complex exponentials

- an advantage in the study of LTI systems :
  A signal can be represented as linear combinations of basic signals.

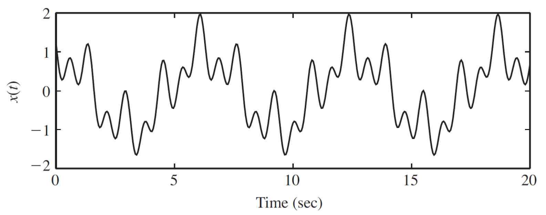
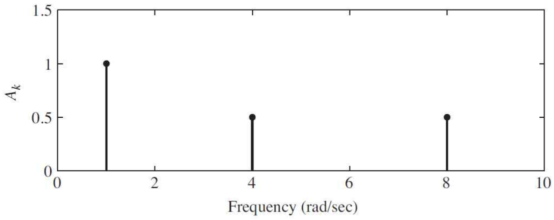
  (a signal) $= \sum$ (constant) $\times$ (basic signal) $\qquad (\triangle)$
- some examples of basic signals

series	basic components (signals)	remarks
Taylor series	$x^0,\ x^1,\ x^2,\ x^3,\ \ldots$	$f(x)=\sum_{n=0}^{\infty}\frac{f^{(n)}(a)}{n!}(x-a)^n$
3-D vector	$(\hat{x},\hat{y},\hat{z}),\ (\hat{r},\hat{\phi},\hat{z}),\ (\hat{r},\hat{\phi},\hat{\theta})$	
complex numbers	$(1,\ j)$	
colors	(red, yellow, blue)	green $= 0$ r $+ 0.5$ y $+ 0.5$ b
sifting	$\delta[n],\ \delta[n\pm 1],\ \delta[n\pm 2],\ \ldots\ldots$	$x[n]=\sum_{k=-\infty}^{\infty}x[k]\delta[n-k]$

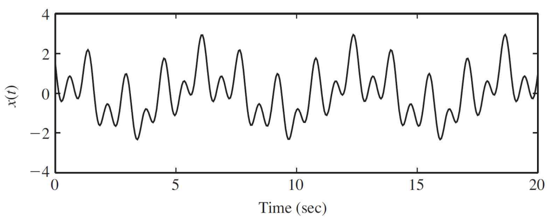
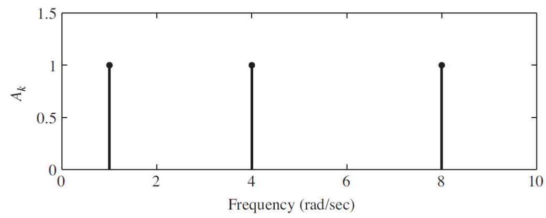
(d) $x(t) = 0.5\cos(t) + 1\cos(4t + \pi/3) + 0.5\cos(8t + \pi/2)$

time domain	frequency domain (Amplitude Spectra)
	

(e) $x(t) = 1\cos(t) + 0.5\cos(4t + \pi/3) + 0.5\cos(8t + \pi/2)$

time domain	frequency domain (Amplitude Spectra)
	

(f) $x(t) = 1\cos(t) + 1\cos(4t + \pi/3) + 1\cos(8t + \pi/2)$

time domain	frequency domain (Amplitude Spectra)
	

- Real Waveform of /ah/

3.3 Fourier series representation of continuous-time periodic signals

§3.3.1 Linear combinations of harmonically related complex exponentials

- For real periodic functions, the Fourier series in terms of complex exponentials is mathematically equivalent to the Fourier series in terms of trigonometric functions.

- The complex exponential form of the Fourier series is particularly convenient for our purpose, so we will use the complex exponential form of the Fourier series from now.

- **Continuous-Time Fourier Series (CTFS) Pair**

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t} \qquad (3.38)$$

$$a_k = \frac{1}{T}\int_T x(t)\, e^{-jk\omega_0 t}\, dt = \frac{1}{T}\int_T x(t)\, e^{-jk(2\pi/T)t}\, dt \qquad (3.39)$$

Eq.(3.38) : synthesis equation
Eq.(3.39) : analysis equation
$\{a_k\}$: Fourier series coefficients or spectral coefficients of $x(t)$
a_0 : DC component of $x(t)$ ($a_0 = \frac{1}{T}\int_0^T x(t)\, dt$) [average value]

(Example 3.5) a periodic square wave

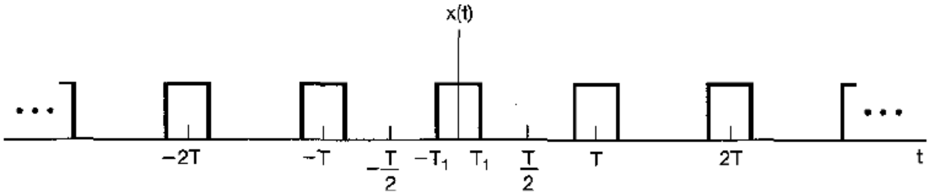


Figure 3.6 Periodic square wave.

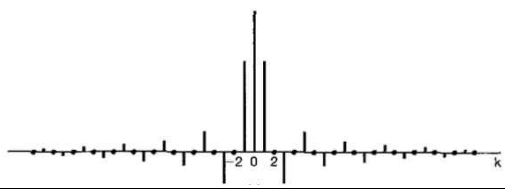
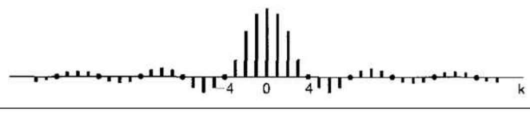
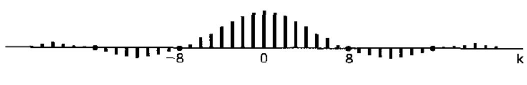
For $k=0$,

$$a_0=\frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}}x(t)\,dt=\frac{1}{T}\int_{-T_1}^{T_1}1\;dt=\frac{2T_1}{T}$$

For $k\neq 0$,

$$\begin{aligned}a_k&=\frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}}x(t)\,e^{-jk\omega_0 t}dt=\frac{1}{T}\int_{-T_1}^{T_1}1\;e^{-jk\omega_0 t}\;dt\\&=-\frac{1}{jk\omega_0 T}\,e^{-jk\omega_0 t}\Big|_{-T_1}^{T_1}\\&=\frac{2}{k\omega_0 T}\left[\frac{e^{jk\omega_0 T_1}-e^{-jk\omega_0 T_1}}{2j}\right]\\&=\frac{2\sin(k\omega_0 T_1)}{k\omega_0 T}=\frac{\sin(k\omega_0 T_1)}{k\pi},\quad k\neq 0\qquad(3.44)\end{aligned}$$

Plots of the Fourier series coefficients Ta_k

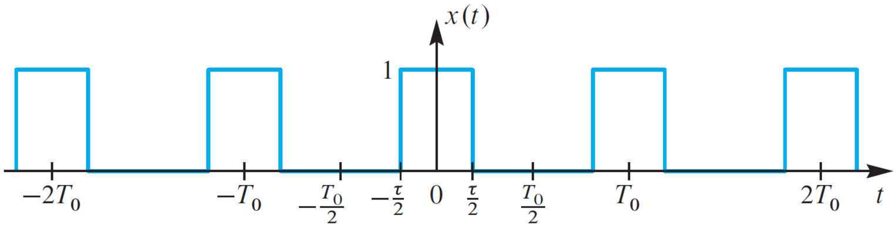
$T=4T_1$ (period)=2(pulse width)	
$T=8T_1$ (period)=4(pulse width)	
$T=16\,T_1$ (period)=8(pulse width)	

the envelope : $(2\sin\omega T_1)/\omega$

the spacing between samples : $2\pi/T$

the first zero-crossing : 2nd / 4th / 8th

[spectrum using the pulse width (τ)]



Fourier coefficients of the pulse train (period $T_0 = T$)

$$a_k = \frac{\tau}{T}\frac{\sin(k\pi\frac{\tau}{T})}{k\pi\frac{\tau}{T}} = \frac{\tau}{T}\,\mathrm{sinc}(k\frac{\tau}{T})$$

Duty cycle ($\frac{\tau}{T}$)	a_k
$\frac{1}{2}$	$\frac{1}{2}\frac{\sin(k\pi\frac{1}{2})}{k\pi\frac{1}{2}} = \frac{\sin(\frac{k\pi}{2})}{k\pi}$
$\frac{1}{4}$	$\frac{1}{4}\frac{\sin(k\pi\frac{1}{4})}{k\pi\frac{1}{4}} = \frac{\sin(\frac{k\pi}{4})}{k\pi}$
$\frac{1}{8}$	$\frac{1}{8}\frac{\sin(k\pi\frac{1}{8})}{k\pi\frac{1}{8}} = \frac{\sin(\frac{k\pi}{8})}{k\pi}$
$\frac{1}{16}$	$\frac{1}{16}\frac{\sin(k\pi\frac{1}{16})}{k\pi\frac{1}{16}} = \frac{\sin(\frac{k\pi}{16})}{k\pi}$

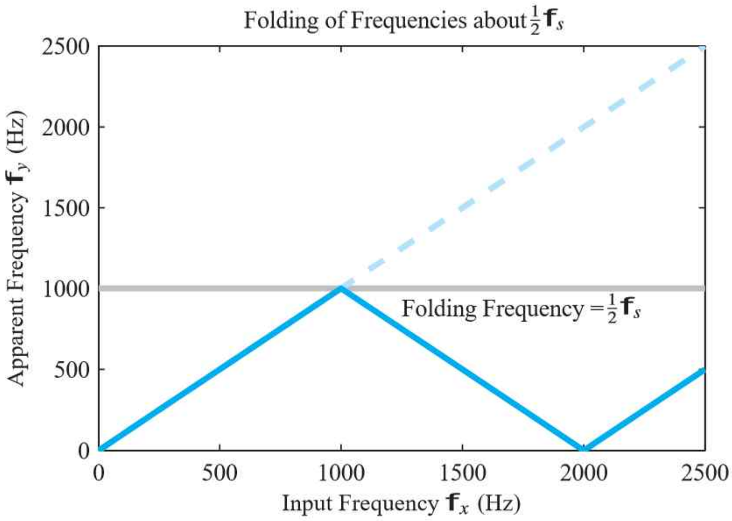
spectrum interval : fundamental frequency $f_0 = \frac{1}{T}$ [Hz]

first zero crossing frequency : $\frac{1}{\tau}$ [Hz] (inverse of the pulse width)

§3.6 Fourier series representation of Discrete-Time Periodic Signal (DTFS)

§3.6.1 Linear Combinations of harmonically related complex exponentials

Fourier series	CTFS	DTFS
periodic function	$x(t) = x(t+T)$	$x[n] = x[n+N]$
frequency	$\omega_0 = 2\pi/T$	$\omega_0 = 2\pi/N$
basic signals	$e^{j\omega_0 t},\ e^{j2\omega_0 t},\ e^{j3\omega_0 t},\ \cdots$	$e^{j\omega_0 n},\ e^{j2\omega_0 n},\ e^{j3\omega_0 n},\ \cdots$
linear combination of basic signals (harmonically related complex exponentials)	$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t}$	$x[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n} = \sum_{k=<N>} a_k e^{jk(2\pi/N)n}$
remark	$e^{j(\omega_0+2\pi)t} \neq e^{j\omega_0 t}$	$e^{j(\omega_0+2\pi)n} = e^{j\omega_0 n}$
summation interval	$-\infty < k < +\infty$ [infinite number of complex exponentials to represent $x(t)$]	$< N >$ [finite number(N) of complex exponentials to represent $x[n]$]



- the Discrete-time Fourier series (DTFS) pair

$$x[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n} = \sum_{k=<N>} a_k e^{jk(2\pi/N)n} \qquad (3.94)$$

$$a_k = \frac{1}{N} \sum_{n=<N>} x[n] e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=<N>} x[n] e^{-jk(2\pi/N)n} \qquad (3.95)$$

※ the continuous-time Fourier series (CTFS) pair

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t} \qquad (3.38)$$

$$a_k = \frac{1}{T} \int_T x(t)\, e^{-jk\omega_0 t}\, dt = \frac{1}{T} \int_T x(t)\, e^{-jk(2\pi/T)t}\, dt \qquad (3.39)$$

Eq.(3.94), (3.38) : the synthesis equations

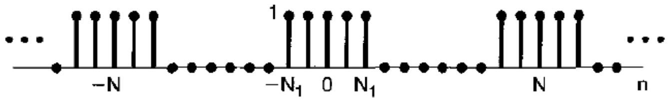
Eq.(3.95), (3.39) : the analysis equations

a_k : the spectral coefficients of $x[n]$ or $x(t)$

※ CTFS vs. DTFS

CTFS	DTFS
$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$	$x[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n}$
INFINITE series	FINITE series

(Example 3.12) Discrete-time periodic square wave (Fig.3.16)



Using Eq.(3.95)

$$a_k = \frac{1}{N}\sum_{n=<N>} x[n]\, e^{-jk(2\pi/N)n}$$

$$= \frac{1}{N}\sum_{n=-N_1}^{N_1} 1\ e^{-jk(2\pi/N)n}$$

Letting $m = n + N_1$

$$a_k = \frac{1}{N}\sum_{m=0}^{2N_1} e^{-jk(2\pi/N)(m-N_1)} \qquad (3.103)$$

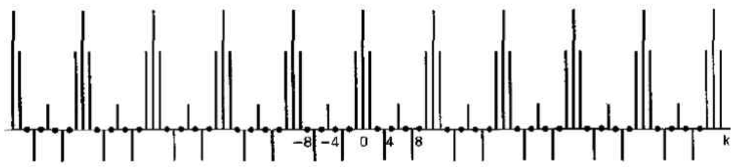
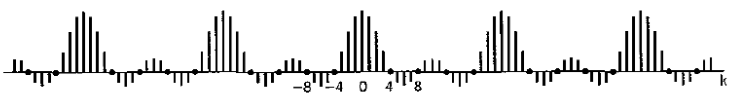
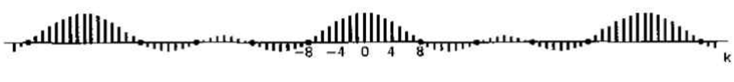
$$= \frac{1}{N} e^{jk(2\pi/N)N_1}\sum_{m=0}^{2N_1} e^{-jk(2\pi/N)m}$$

$$= \frac{1}{N} e^{jk(2\pi/N)N_1}\left(\frac{1-e^{-jk2\pi(2N_1+1)/N}}{1-e^{-jk(2\pi/N)}}\right)$$

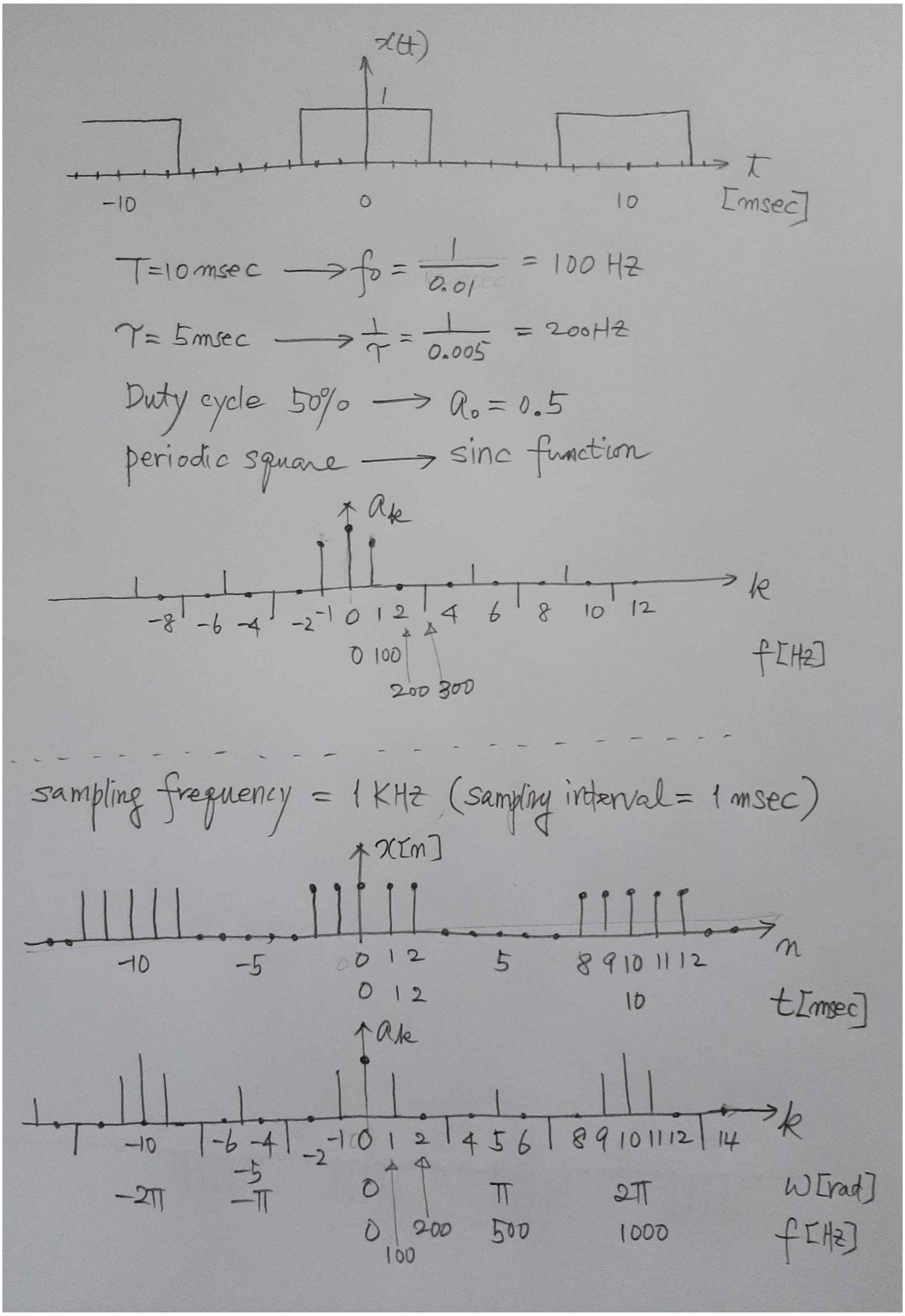
$$= \frac{1}{N}\ \frac{e^{-jk(2\pi/2N)}\left[e^{jk2\pi(N_1+1/2)/N} - e^{-jk2\pi(N_1+1/2)/N}\right]}{e^{-jk(2\pi/2N)}\left[e^{jk(2\pi/2N)} - e^{-jk(2\pi/2N)}\right]}$$

$$= \frac{1}{N}\ \frac{\sin\left[2\pi k(N_1+1/2)/N\right]}{\sin(\pi k/N)},\quad k \neq 0, \pm N, \pm 2N, \cdots \qquad (3.104)$$

and $a_k = \dfrac{2N_1+1}{N},\quad k = 0, \pm N, \pm 2N, \cdots$ (3.105)

a_k for $2N_1+1=5$ and $N=10$	
a_k for $2N_1+1=5$ and $N=20$	
a_k for $2N_1+1=5$ and $N=40$	

- Comparison CTFS / DTFS (sampling)



Summary

● Continuous-Time Fourier Series (CTFS) Pair

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t} \qquad (3.38)$$

$$a_k = \frac{1}{T}\int_T x(t)\, e^{-jk\omega_0 t}\, dt = \frac{1}{T}\int_T x(t)\, e^{-jk(2\pi/T)t}\, dt \qquad (3.39)$$

● Discrete-Time Fourier Series (DTFS) pair

$$x[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n} = \sum_{k=<N>} a_k e^{jk(2\pi/N)n} \qquad (3.94)$$

$$a_k = \frac{1}{N}\sum_{n=<N>} x[n]\, e^{-jk\omega_0 n} = \frac{1}{N}\sum_{n=<N>} x[n]\, e^{-jk(2\pi/N)n} \qquad (3.95)$$

● Continuous-Time Fourier Transform (CTFT) Pair

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t} d\omega \qquad (4.8)$$

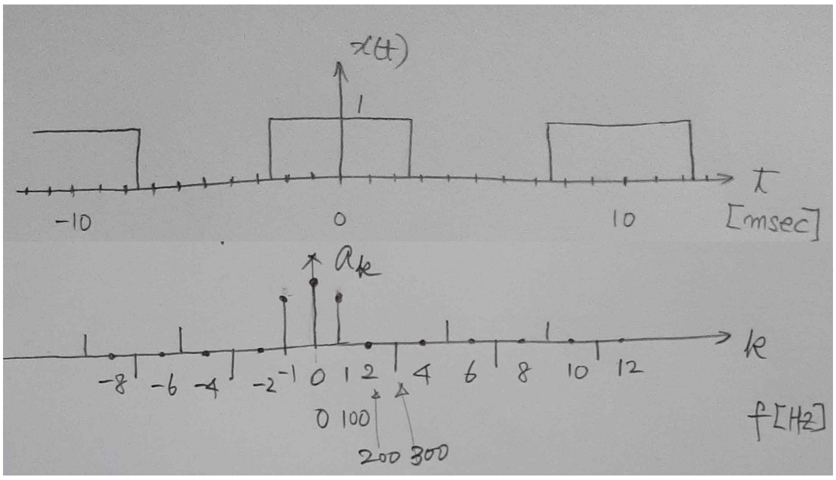
$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t} dt \qquad (4.9)$$

● Discrete-Time Fourier Transform (DTFT) Pair

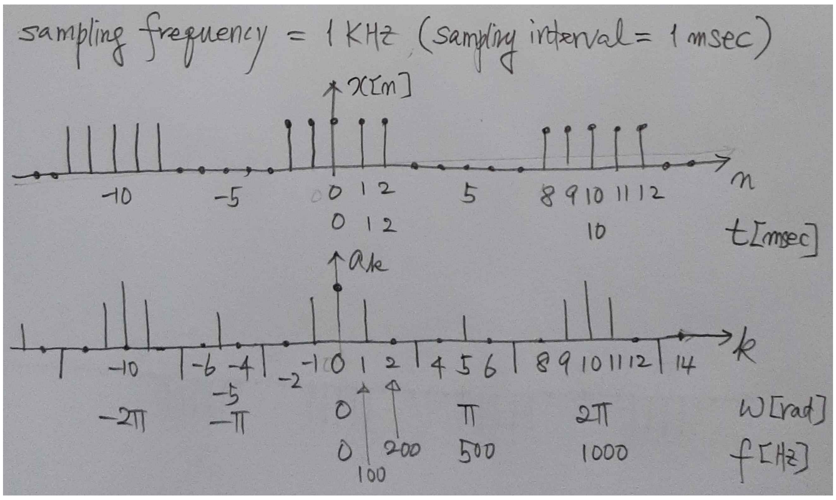
$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega \qquad (5.8)$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n]\, e^{-j\omega n} \qquad (5.9)$$

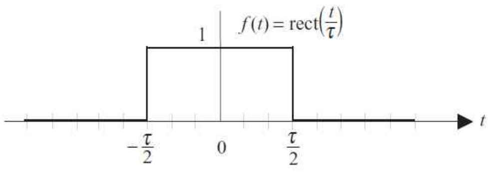
CTFS

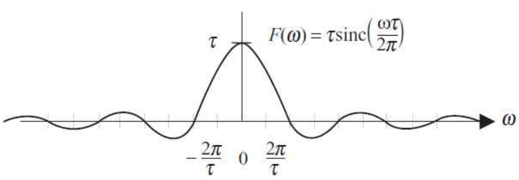


DTFS

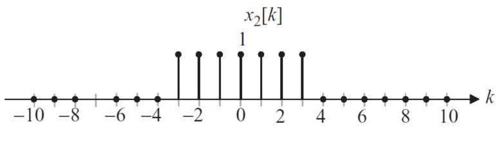


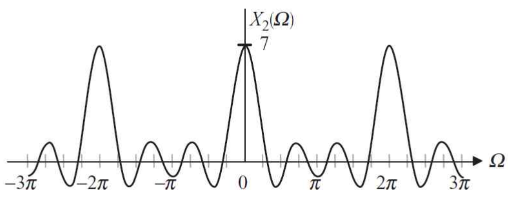
CTFT





DTFT



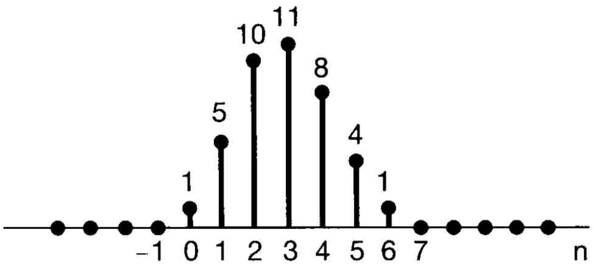


● Conclusion in this semester

- (Chapter 2) Sifting Property in TIME DOMAIN

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

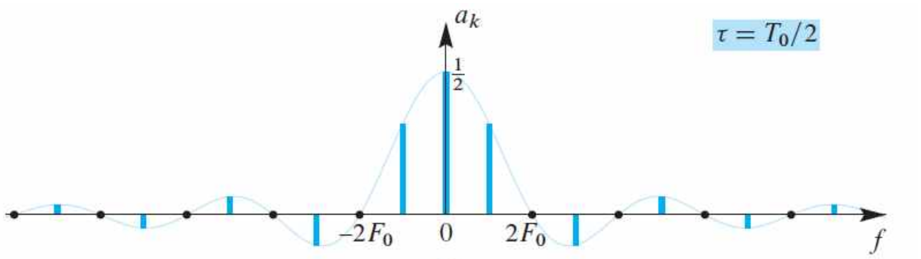
(Example)



$$\begin{aligned} x[n] &= x[0]\delta[n] + x[1]\delta[n-1] + x[2]\delta[n-2] + x[3]\delta[n-3] \\ &\quad + x[4]\delta[n-4] + x[5]\delta[n-5] + x[6]\delta[n-6] \\ &= 1\,\delta[n] + 5\,\delta[n-1] + 10\,\delta[n-2] + 11\,\delta[n-3] \\ &\quad + 8\,\delta[n-4] + 4\,\delta[n-5] + 1\,\delta[n-6] \end{aligned}$$

- (Chapter 3) Sifting Property in FREQUENCY DOMAIN

(Example) **Fourier Series**



$$\begin{aligned} X[k] &= \cdots + a_{-3}\,\delta[k-(-3)] + a_{-2}\,\delta[k-(-2)] + a_{-1}\,\delta[k-(-1)] \\ &\quad + a_0\,\delta[k] + a_1\,\delta[k-1] + a_2\,\delta[k-2] + a_3\,\delta[k-3] + \cdots \\ X(\omega) &= \cdots + a_{-3}\,\delta[\omega + 3\omega_0] + a_{-2}\,\delta[\omega + 2\omega_0] + a_{-1}\,\delta[\omega + \omega_0] \\ &\quad + a_0\,\delta[\omega] + a_1\,\delta[\omega - \omega_0] + a_2\,\delta[\omega - 2\omega_0] + a_3\,\delta[\omega - 3\omega_0] + \cdots \end{aligned}$$

※ (Chapter 2) A signal is a linear combination of delta functions in TIME DOMAIN.

(Chapter 3) A signal is a linear combination of delta functions in FREQUENCY DOMAIN. [Fourier Series]

(Chapter 2 & 3) A signal is a linear combination of delta functions in TIME and FREQUENCY DOMAINs. [sifting property]